On the Investment Network and Development

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Motivation

Salient features of the process of economic development:

Capital accumulation **Productivity growth** **Sectorial reallocation**

sectors use different capital types → produced by other sectors or imported

Investment Network

Hirschman ('58)'s hypothesis: investment linkages important for economic development

Motivation

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Investment Network

Hirschman ('58)'s hypothesis: investment linkages important for economic development

What is the role of investment network for income disparities?

This paper

1. **Model**: dynamic multisector open economy + **investment network** + IO
   - GDP elasticity to sectoral productivity = $\underbrace{\text{on impact}}_{\text{Hulten}} + \underbrace{\text{capital deepening}}_{\text{transition+LR}}$

2. New harmonized estimates of **cross-country investment networks**
   - Methodology: ≈ BEA in the US
   - Coverage: 58 countries w/income per capita \$428 and \$81599 constant ppp USD
     9 from sSA; 20 with time series from 1960s.

3. **Sectoral influence** and its channels along the development spectrum

4. Role of the investment network for **income disparities**
   - Development accounting
   - Transition dynamics (Korea 1965–2014)

Main findings

Theory:

- **1SGM mis-specified** when investment technology ≠ consumption goods technology
  - MPK depends on productivity ⟶ can't separate technology from capital
- **Sectoral influence**: $\frac{d\ln(GDP)}{d\ln(a)}$ = impact + capital deepening + gross return on capital
  - Domar weights depend on investment rates ⟶ on the whole transition path

Quantitatively:

- **CONS & DUR**: influence via capital deepening, hump-shaped;
  ICT rises w/ income through all channels.
- *Accounting*: Ω disparities account for ≈40% of income variance.
- *Transitions:* relative prices ≠ Δ investment network Korea's '65–'14.
  Gross return pass-through important for ICT and Manufacturing

Contribution

1. Sectoral influence with durable goods: on impact vs. capital deepening; Domar weights depend on investment dynamics

   non-durables: Hulten (1978); Acemoglu et.al. (2012), Baqaee & Fahri (2020), Liu (2019), Bigio & La'O (2020)
   long-run comparisons: Baqaee & Malmberg (2025)
   short/medium-term:: vom Lehn & Winberry (2021) & Ho et.al. (2026)

2. First measures of sectoral investment demand along the development spectrum

   - Evidence of shifts in the composition of investment Garcia-Santana, et.al. 2021; Herrendorf, et.al. 2021
   - Shifts in the demand for investment and (VA) structural change Caunedo & Keller, 2023

   US and in OECD countries → not harmonized vomLehn & Winberry, 2021 and Ding, 2025
   Firm-to-firm transactions (Chile): only domestic Gillmore et. al., 2025

3. Role of the investment network for cross-country income differences: Buera & Trachter (2024)

   IO linkages: Ciccone (2002); Jones (2011); Fadinger, et.al. (2022)

Outline

1. Simple (toy) economy
2. Multisector model: channels of sectoral influence
3. Measurement of the investment network
4. Sectoral influence along the income spectrum
5. Implications for economic development

Fixing ideas

A simple economy

A simple economy

Two sectors, no intermediate inputs, no trade, same output elasticities

- Sectoral technologies

$$y_{it} = z_i K_{it}^{\alpha} L_{it}^{1-\alpha} \qquad z_1 \neq z_2$$

$$y_{it} = c_{it} + \overbrace{\sum_j x_{ijt}}^{x_{it}}$$

A simple economy

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$$v_{it} = c_{it} + \overbrace{\sum_j x_{ijt}}^{x_{it}}$$

- Capital accumulation

$$K_{it+1} = (1-\delta)K_{it} + \prod_j x_{jit}^{\omega_j}$$

implication: same investment aggregator $\rightarrow p_i^k = p_j^k \rightarrow \frac{K_i}{L_i} = \frac{K}{L}$

- GDP: $\ln(v_t) = \sum_i \tilde{\eta}_{it} \ln(v_{it})$ for $\tilde{\eta}_i$ the expenditure share of sector i

A simple economy: steady state

Two sectors, no intermediate inputs, no trade, same output elasticities

Output per worker

$$\ln(v) = \frac{1}{1-\alpha} \sum_{i=1,2} \tilde{\eta}_i \left(\ln z_i + \ln(\tilde{\eta}_i)\right) + \frac{\alpha}{1-\alpha} \ln\left(\frac{K}{v}\right)$$

where is the investment network?

A simple economy: steady state

Two sectors, no intermediate inputs, no trade, same output elasticities

Output per worker

$$\ln(v) = \frac{1}{1-\alpha} \sum_{i=1,2} \tilde{\eta}_i \left(\ln z_i + \ln(\tilde{\eta}_i)\right) + \frac{\alpha}{1-\alpha} \ln(\frac{K}{v})$$

where is the investment network? 1SGM ($\omega = \tilde{\eta}$):

$$\text{MPK} \equiv \alpha \frac{v}{K}$$

A simple economy: steady state

Two sectors, no intermediate inputs, no trade, same output elasticities

Output per worker

$$\ln(\nu) = \frac{1}{1-\alpha} \sum_{i=1,2} \tilde{\eta}_i \left(\ln z_i + \ln(\tilde{\eta}_i)\right) + \frac{\alpha}{1-\alpha} \ln\left(\frac{K}{\nu}\right)$$

where is the investment network? General ($\omega \neq \tilde{\eta}$):

$$\text{MPK} \equiv \alpha \frac{\nu}{K} \prod_j z_j^{\omega_j - \tilde{\eta}_j} \prod_j \frac{\omega_j^{\omega_j}}{\tilde{\eta}_j^{\tilde{\eta}_j}}$$

- $\omega_j > \tilde{\eta}_j$ (e.g. construc.): $z_j \uparrow \Rightarrow \frac{K}{\nu} \uparrow$ $\quad$ $\omega_j < \tilde{\eta}_j$ (e.g. services): $z_j \uparrow \Rightarrow \frac{K}{\nu} \downarrow$

A simple economy: steady state

Two sectors, no intermediate inputs, no trade, same output elasticities

Output per worker

$$\ln(\nu) = \frac{1}{1-\alpha} \sum_{i=1,2} \tilde{\eta}_i \left(\ln z_i + \ln(\tilde{\eta}_i)\right) + \frac{\alpha}{1-\alpha} \ln\left(\frac{K}{\nu}\right)$$

solving for equilibrium $\frac{K}{\nu}$

$$\ln(\nu) = \sum_j \Bigg(\underbrace{\tilde{\eta}_j}_{\text{on impact}} + \underbrace{\frac{\alpha\,\omega_j}{1-\alpha}}_{\text{capital deepening}} \Bigg) \ln z_j + \text{const.}$$

- without heterogeneity in factor shares $\omega_j \to \neq \frac{\partial \ln(\nu)}{\partial \ln(z)}$, but no aggregate effect.
- with heterogeneous $\alpha \to \Omega$ matters

Multisector Model

of Investment Linkages

The Model

- N sectors, CRS technologies in capital k, labor l and intermediate inputs m

$$\underbrace{y_{nt}}_{\text{gross output}} = G_n(v_{nt}, m_{nt}), \qquad \underbrace{v_{nt}}_{\text{value added}} = z_{nt} V_n(k_{nt}, l_{nt}), \qquad m_{nt} = M_n(m_{1nt}, \ldots, m_{Nnt})$$

  shares: γ_n VA in gross output, α_n capital in VA, $\mu_{in} \rightarrow$ IO matrix M

- Sectors produce for final uses c, intermediates m and domestic investment, χ^d

$$y_{nt} = c_{nt} + \sum_i m_{nit} + \sum_i \chi^d_{nit}$$

- Non-Homothetic CES final output aggregator

$$\sum_{n=1}^{N} \theta_n^{\frac{1}{\sigma}} (c_{nt})^{\frac{\sigma-1}{\sigma}} Y_t^{\frac{\epsilon_n - 1}{\sigma}} = Y_t^{\frac{\sigma-1}{\sigma}}, \qquad Y_t = C_t + \underbrace{e_t}_{\text{exports}}$$

- Small open economy: exports final goods for imported investment goods; balanced trade

The Model

- Sector-specific capital with standard law of motion

$$k_{nt+1} = x_{nt} + (1 - \delta_n)k_{nt}$$

- CRS investment aggregator

$$x_{nt} = X_n(\chi_{1nt}, \ldots, \chi_{Nnt}) \quad \rightarrow \quad \text{Investment network, } \Omega, \text{ w/ } \omega_{in} \equiv \frac{p_i^\chi \chi_{in}}{p_n^x x_n}, \ \sum_i \omega_{in} = 1$$

quantitative: CES, $\omega_{in} = \tilde{\omega}_{in} \left(\frac{p_i^\chi}{p_n^x} \right)^{1-\iota}$

- Investment goods produced domestically or imported

$$\chi_{int} = \underbrace{\left(\frac{\chi_{int}^d}{1-\phi_i}\right)^{1-\phi_i}}_{\text{domestic}} \underbrace{\left(\frac{\chi_{int}^f}{\phi_i}\right)^{\phi_i}}_{\text{imported}}$$

terms of trade: $\mathcal{T}_i \equiv \frac{p}{p_i^f}$

AGP trade

Domar weights

Proposition (Domar weights)

$$\tilde{\eta}_t = \Big[I - \underbrace{(I-\phi)\Omega_t \Gamma_t \alpha_t \rho_t}_{\textit{investment links}} - \underbrace{M_t(I-\Gamma_t)}_{\textit{IO links}} \Big]^{-1} \underbrace{\tilde{\zeta}_t}_{\textit{final uses}} \qquad \rho_{nt} \equiv \frac{p^x_{nt} x_{nt}}{R_{nt} k_{nt}} = \frac{x_{nt}/k_{nt}}{r^x_{nt} + \delta_n}$$

- ρ: investment expenditure relative to capital income
- Domar weights depend on investment rates $\longrightarrow$ on the entire future path

  novel to durable inputs; absent with non-durable inputs
- All the dynamics is in $\rho_{nt} \equiv \frac{p^x_{nt} x_{nt}}{R_{nt} k_{nt}} = \frac{k_{nt+1}/k_{nt} - (1-\delta_n)}{r^x_{nt} + \delta_n} = \frac{g^k_t + \delta_n}{r^x_{nt} + \delta_n}$
- At the Phelps golden rule, $r^x_{nt} = g^k_t$ the network is irrelevant!

welfare

Impact of productivity on GDP

Proposition (GDP)

$$\ln(\nu_t) = \underbrace{\Phi_t \eta_t}_{\text{sectoral influence}} \left[\Gamma_t \ln z_t + \underbrace{\Gamma_t \alpha_t \Omega_t' \phi \ln \mathcal{T}_t}_{\text{terms of trade}} + \underbrace{\Gamma_t \alpha_t (\ln \rho_t - \ln \xi_t)}_{-\alpha \ln(R/p^x)} \right] + \underbrace{\Phi_t \rho_t}_{\text{curvature}}$$

Impact of productivity on GDP

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$$\eta_t = \underbrace{\zeta_t'}_{\text{final-demand share}} \underbrace{\left[I - \Gamma_t \alpha_t \Omega_t' (I - \phi) - (I - \Gamma_t) M_t' \right]^{-1}}_{\text{augmented Leontief inverse, } \Xi_t} \qquad \underbrace{\Phi_t \equiv \left(1 - \eta_t \Gamma_t \alpha_t \Omega_t' \phi \mathbf{1} \right)^{-1}}_{\text{trade amplification}}$$

for z sectoral TFP, α capital share, Γ VA share, $(1 - \phi)$ domestic investment share, $\xi = x/k$; ρ: curvature & labor share

Influence vs. Domar weights

$$\underbrace{\eta_t}_{\text{influence}} = \underbrace{\tilde{\eta}_t'}_{\text{Domar}} + \underbrace{\tilde{\eta}_t' \Gamma_t \alpha_t \Omega_t' (I - \phi) \Xi_t}_{\text{productivity built into others' capital}}$$

- Influence ≠ Domar weight because capital is produced
- Investment network enters through capital deepening only
- Large for sectors that supply capital to large, capital-intensive buyers

Channels of sectoral influence

$$\ln(\nu_t) = \underbrace{\tilde{\eta}_t' \Gamma_t \ln z_t}_{\text{on impact}} + \underbrace{\left[\overbrace{\tilde{\eta}_t' \Gamma_t \alpha_t \Omega_t' (I - \phi) \Xi_t \Gamma_t}^{\text{domestic } \Delta k} + \overbrace{(\Phi_t - 1)\eta_t \Gamma_t}^{\text{imported } \Delta k}\right] \ln z_t + \overbrace{\text{TOT}_t}^{\text{terms of trade}}}_{\text{capital deepening}}$$

$$+ \underbrace{\Phi_t \eta_t \Gamma_t \alpha_t (\ln \rho_t - \ln \xi_t)}_{\text{transition}} + \underbrace{\Phi_t \rho_t}_{\text{curvature, labor share}}$$

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Channels of sectoral influence, sector j: domestic Δk

$$\underbrace{\sum_i \tilde{\eta}_i \gamma_i \alpha_i}_{i\text{'s capital income}} \sum_l \underbrace{\omega_{li}}_{l \text{ in } i\text{'s invest.}} \underbrace{(1-\phi_l)}_{\text{domestic}} \underbrace{\Xi_{lj}}_{z_j \text{ in cost of } l}$$

- $z_j \uparrow \rightarrow$ goods l cheaper (directly if $l = j$, or via IO & investment chains) $\rightarrow$ buyers i that invest in l accumulate k
- Direct term ($\Xi_{lj} \approx \mathbf{1}\{l = j\}$):

$$\gamma_j (1-\phi_j) \sum_n \tilde{\eta}_n \gamma_n \alpha_n \, \omega_{jn}$$

  $\rightarrow$ j as a capital supplier, weighted by buyers' capital income

- Toy economy (steady states, $\gamma = 1$, $\phi = 0$, common ω): $\dfrac{\alpha\,\omega_j}{1-\alpha}$

Channels of sectoral influence, sector j: transition

$$\underbrace{\Phi_t \sum_i \eta_{it}\gamma_{it}\alpha_{it}\left(\ln \rho_{it} - \ln \xi_{it}\right)}_{\text{transition: gross return on capital}} = -\Phi_t \sum_i \eta_{it}\gamma_{it}\alpha_{it} \ln\left(\frac{R_{it}}{p^x_{it}}\right)$$

- k_{it} predetermined $\rightarrow$ gross return jumps, offsets capital deepening on impact
- Falls back as k accumulates; $\rightarrow$ constant along the AGP.
- Gross return

$$\frac{R_{it}}{p^x_{it}} = r^x_{it} + \delta_i = \frac{\xi_{it}}{\rho_{it}}, \qquad \xi_{it} = g^k_{it} + \delta_i$$

- No arbitrage:

$$1 + r^x_{i,t+1} = (1 + r_{t+1})\frac{p^x_{it}}{p^x_{i,t+1}}$$

Cross-Country Investment Networks

Methodology

Methodology

Use Tables (World IO Database): capital produced or imported

How is capital produced/imported by a sector purchased by other sectors?

- Imputation approach ≈ BEA's methodology in the US:
  1. manual allocation
  2. proportional to occupational composition of the sector
  3. proportional to capital expenditures (handful of sectors w/microdata)

Equipment Sectors

- Tools used on the job (Caunedo et.al.,'23)

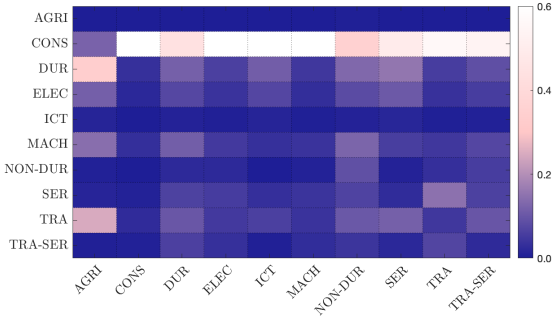
Other Sectors

- Assign capital proportional to intermediate inputs

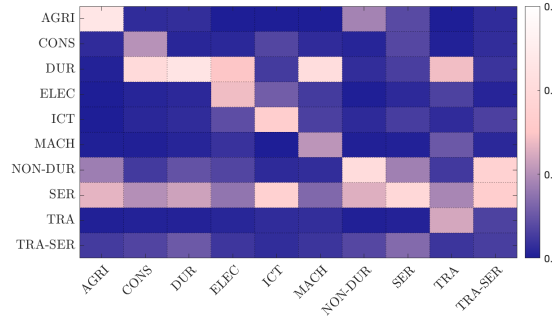
example data coverage

Investment Network vs IO Network 2014

INV Network: India



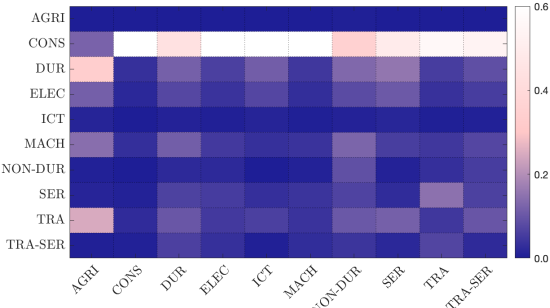
IO Network: India



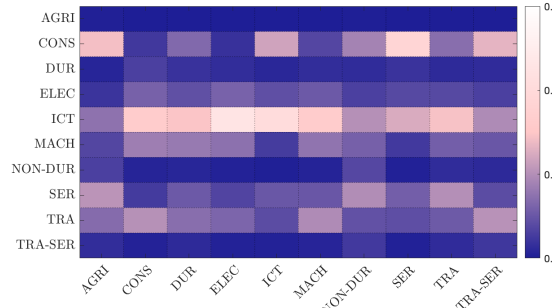
details

Investment Network along Development

INV Network: India



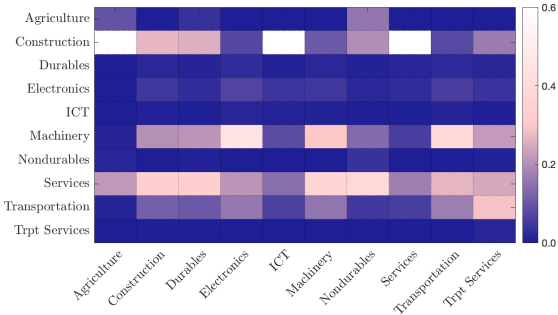
INV Network: USA



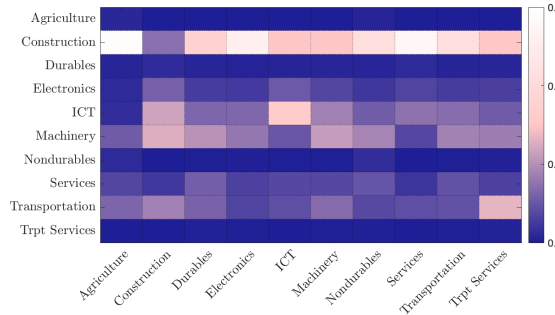
validation vLW | relative prices | validation Chile | outdegrees | HHI demand

Investment Network along Development: South Korea

1965 (GDP p/cap. 1450)



2014 (GDP p/cap. 35524)

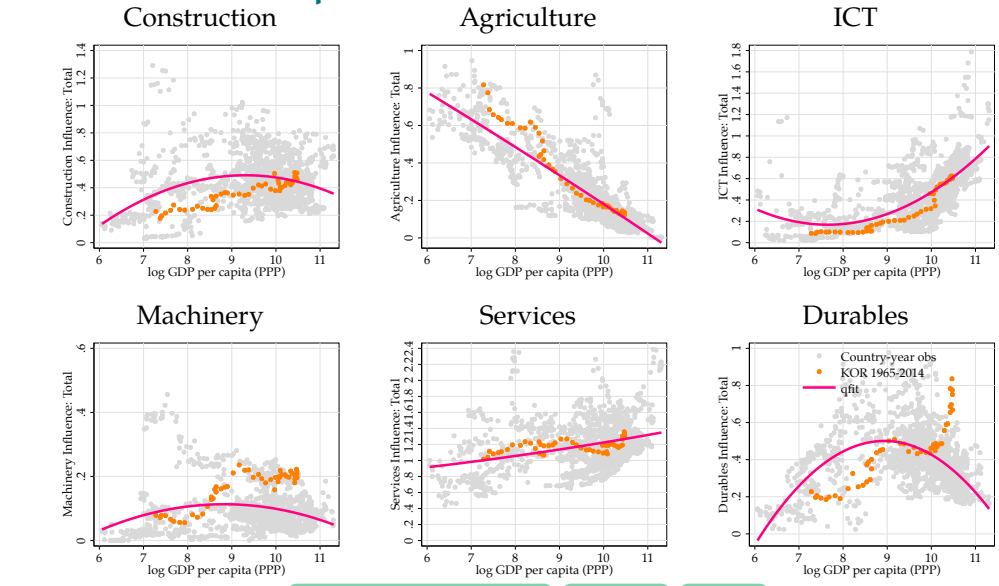


CONS prominent throughout; ICT↑, SERV↓; bundles diversify with income

Sectoral Influence

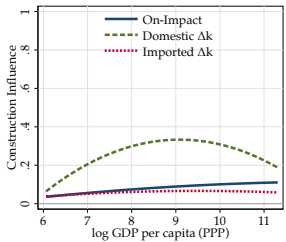
along the development spectrum

Sectoral Influence: $\Phi\eta\Gamma$

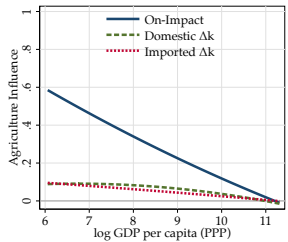


Channels of Sectoral Influence along Development

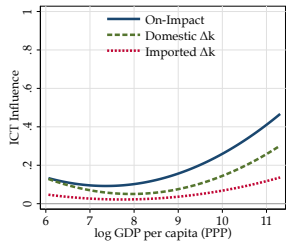
Construction



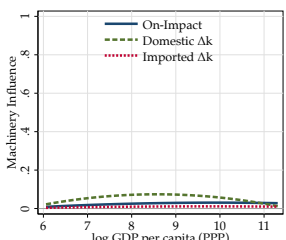
Agriculture



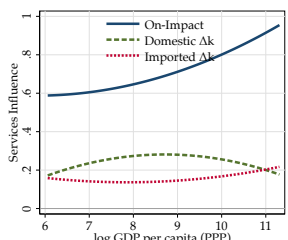
ICT



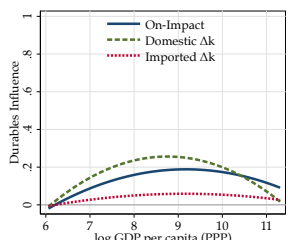
Machinery



Services



Durables



other sectors

Channels of Sectoral Influence: Takeaways

- **CONS**: influence is mostly domestic capital deepening; hump-shaped, peaks at middle income
- **DUR**: capital deepening ≈ on impact; both hump-shaped
- **ICT**: rises with income through all three channels; imported Δk grows at high income
- **AGRI**: on impact only, falls with development (structural change)
- **SERV**: on impact dominates and rises; sizable capital deepening (domestic & imported)

→ capital deepening is where Ω matters: sectors' role as capital suppliers, not their size

Development Accounting

Quantitative Exercises

1. Infer sectoral productivity-like shifters to match observed sectoral value added,
   $\mathfrak{a} = (\Phi\Xi)^{-1} \ln(\mathbf{v}), \quad \mathfrak{a} \equiv \Gamma \ln \mathbf{z} + \Gamma\alpha\Omega'\phi \ln \mathcal{T}$

2. To interpret residuals as productivity:
   common technology in a sector across countries + Hicks neutral productivity.

$$\ln(v_j^c) = \ln z_j^c + h_j(\ln k_j^c, \ln l_j^c, \ln m_j^c)$$

3. How much of the disparities in income per capita come from Ω?

   - when model-predicted output disparities come only from Ω
   - when sectoral productivity interact with Ω
   - when capital intensity is the same across countries
   - when only domestic investment flows are considered

What is the role of Ω for income variance?

Table 1: Development Accounting

	Income Variance	Contribution of Ω
Baseline	1	
Investment Links	0.42	42%

$$\ln(\boldsymbol{\nu}) = \underbrace{\Phi}_{\substack{\text{trade} \\ \text{amplification}}} \quad \underbrace{\zeta'}_{\text{final-demand share}} \quad \underbrace{(I - \Gamma\alpha\Omega'(I - \phi) - (I - \Gamma)M')^{-1}\,\Gamma\mathbf{a}}_{\substack{\text{augmented} \\ \text{Leontief inverse}}}$$

$\longrightarrow$ **42%** of GDP p/cap. variance explained by $\neq$ in the investment network

What is the role of Ω for income variance?

Table 2: Development Accounting

	Income Variance	Contribution of Ω
Baseline	1	
Investment Links	0.42	42%
Investment Links + productivity$_{US}$	0.41	41%

$$\ln(\nu) = \underbrace{\Phi}_{\text{trade amplification}} \quad \underbrace{\zeta'}_{\text{final-demand share}} \quad \underbrace{(I - \Gamma\alpha\Omega'(I-\phi) - (I-\Gamma)M')^{-1}}_{\text{augmented Leontief inverse}} \Gamma a_{US}$$

$\rightarrow$ $\approx$**98%** of investment links contribution explained by $\neq$ in influence

What is the role of Ω for income variance?

Table 3: Development Accounting

	Income Variance	Contribution of Ω
Baseline	1	
Investment Links	0.42	42%
Investment Links + productivity$_{US}$	0.41	41%
Investment Links + α avg.	0.62	62%

$$\ln(\boldsymbol{\nu}) = \underbrace{\Phi}_{\substack{\text{trade} \\ \text{amplification}}} \quad \underbrace{\zeta'}_{\text{final-demand share}} \quad \underbrace{(I - \Gamma\alpha\Omega'(I - \phi) - (I - \Gamma)M')^{-1}}_{\substack{\text{augmented} \\ \text{Leontief inverse}}} \Gamma\mathbf{a}$$

$\longrightarrow$ Δ Capital-intensity dampens the role of Ω.

What is the role of Ω for income variance?

Table 4: Development Accounting

	Income Variance	Contribution of Ω
Baseline	1	
Investment Links	0.42	42%
Investment Links + productivity$_{US}$	0.41	41%
Investment Links + α avg.	0.62	62%
Only Domestic Investment Links	1.04	–

$$\ln(\nu) = \underbrace{\Phi}_{\text{trade amplification}} \quad \underbrace{\zeta'}_{\text{final-demand share}} \quad \underbrace{(I - \Gamma\alpha\Omega^{Dom\prime} - (I-\Gamma)M')^{-1}}_{\text{augmented Leontief inverse}} \Gamma a$$

→ Trade in capital goods: mild force for convergence (variance +4%).

avg. GDP p/cap. falls 20% without capital-goods imports

income levels, no trade | full table

General Equilibrium

sectorial productivity growth & the investment network

Transition Dynamics: South Korea 1965–2014

- **Goal**: quantify role of sectorial productivity & Ω for output and capital accumulation
- 5 sectors: AGRI, CONS, ICT, MAN, SERV
- CES investment aggregator; elasticity ι estimated from Korean data, 1965–2014:

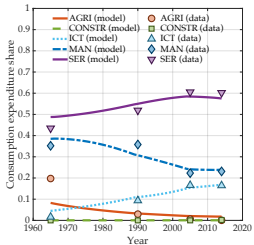
$$\ln\left(\frac{\omega_{int}}{\omega_{\text{MAN},nt}}\right) = \underbrace{\ln\left(\frac{\tilde{\omega}_{in}}{\tilde{\omega}_{\text{MAN},n}}\right)}_{\text{FE}_{in}} + (1-\iota)\ln\left(\frac{p_{it}}{p_{\text{MAN},t}}\right) + \varepsilon_{int} \quad \Rightarrow \quad \iota = 2.41$$

- loadings $\tilde{\omega}_{in,t}$ chosen so model & data investment expenditure shares coincide
- productivities calibrated to match relative prices & GDP per capita growth
- calibrated at four anchor years (1966, 1990, 2005, 2014); interpolated in between

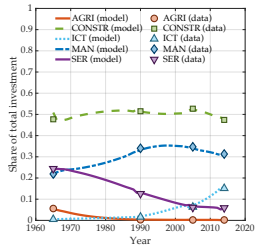
calibration transition paths

Model Fit: Targeted Moments

Sectoral Consumption Shares



Sectoral Investment Shares

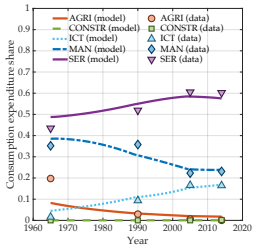


Model (lines) vs data (markers).

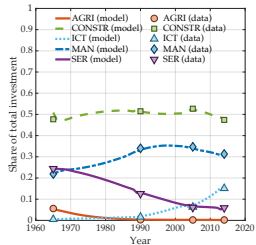
investment network by destination

Model Fit: Targeted Moments

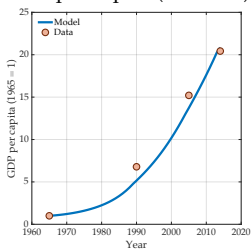
Sectoral Consumption Shares



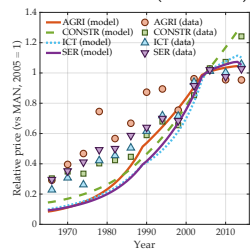
Sectoral Investment Shares



GDP per capita (1965 = 1)



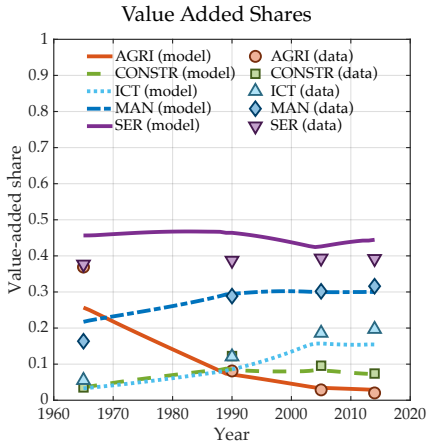
Relative Prices (2005 = 1)

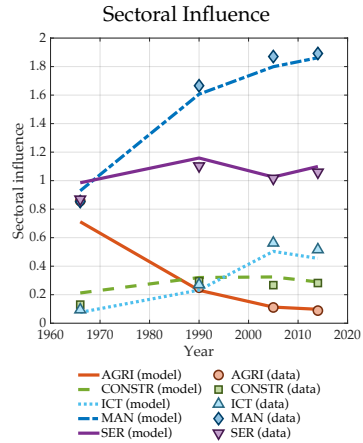


Model (lines) vs data (markers).

investment network by destination

Untargeted Moments: VA Shares & Sectoral Influence

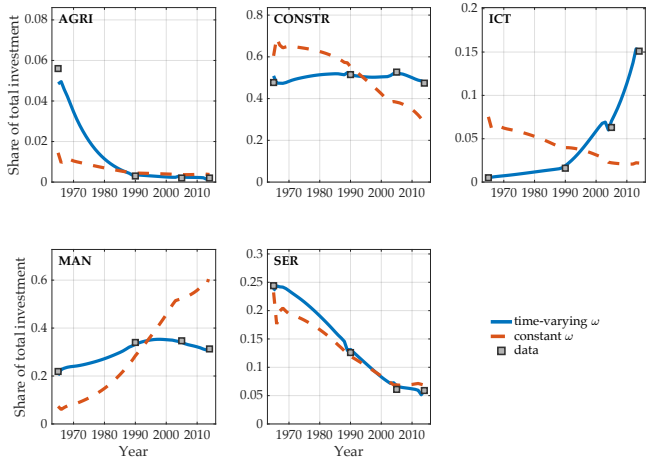




- MAN influence doubles ($\approx 0.9 \to 1.8$); ICT rises ($<0.2 \to \approx 0.6$); AGRI falls ($\approx 0.7 \to 0.1$)

transition paths: levels

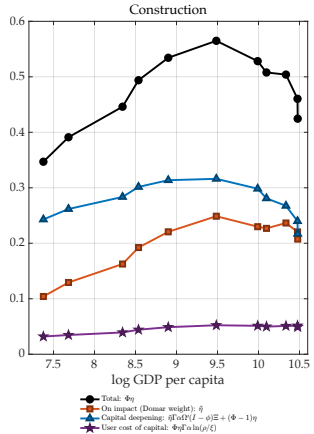
Counterfactual: the Role of Relative Prices for Changes in the Invest. Network



- Constant $\tilde{\omega}$: prices capture the fall of SERV ($\approx$25% $\rightarrow$ 5%), miss rise of ICT & CONS
- MAN share too low early (7% vs 22%) and too high late (60% vs 31%)

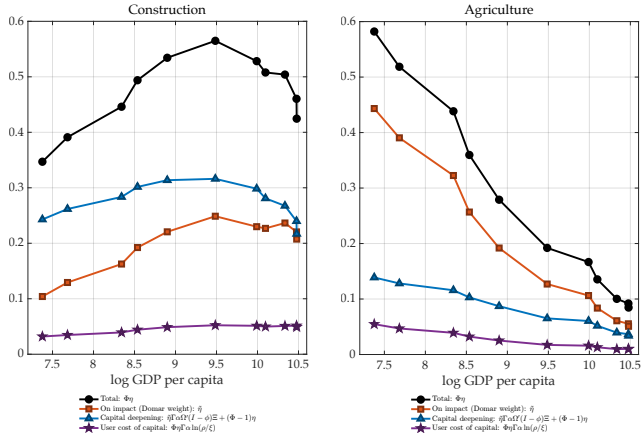
Investment network, constant $\tilde{\omega}$

Channels of Sectoral Influence along Korea's Transition



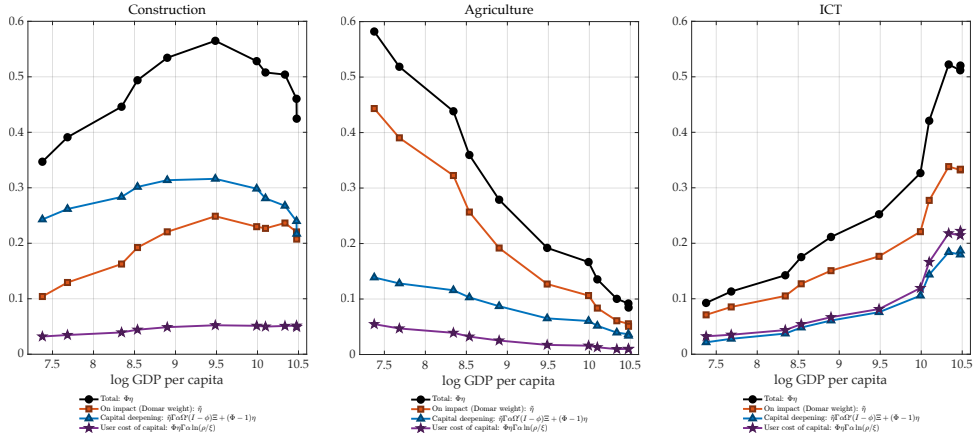
MAN & SERV | shares of total

Channels of Sectoral Influence along Korea's Transition



MAN & SERV | shares of total

Channels of Sectoral Influence along Korea's Transition



MAN & SERV | shares of total

Counterfactual: the Role of the Investment Network

Scenario	$\Delta \ln Y$	$\Delta \ln(K/Y)$	$\frac{\Delta \ln Y}{\Delta \ln(K/Y)}$	$\Delta \ln Y - \frac{\bar{\alpha}}{1-\bar{\alpha}} \Delta \ln(K/Y)$	productivity % of $\Delta \ln Y$
Baseline	2.04	0.88	2.31	1.66	81%
Constant $\tilde{\omega}$	1.97	0.69	2.84	1.67	85%
Ω = KOR 1965	1.93	1.10	1.75	1.45	75%
Ω = value-added shares	1.92	0.87	2.20	1.54	80%
Ω = identity	2.04	0.69	2.98	1.74	85%

South Korea, 1965–2014. Detrended log changes, chained Törnqvist Y and K. Ω = VA shares uses a Cobb–Douglas investment aggregator ($\iota = 1$); others use $\iota = 2.41$. $\bar{\alpha}$: simple average of sectoral capital shares.

Illustrative economy (steady state): $$\frac{\Delta \ln \nu}{\Delta \ln(K/\nu)} = \frac{\alpha}{1-\alpha} + \frac{1}{1-\alpha} \frac{\sum_j \tilde{\eta}_j \, \Delta z_j}{\sum_j (\omega_j - \tilde{\eta}_j) \, \Delta z_j}$$

→ **Changes in the investment network** explain 24% ($= 1 - 1.75/2.31$)

For comparison, at the average capital share $\bar{\alpha} = 0.30$: $\frac{\bar{\alpha}}{1-\bar{\alpha}} = 0.44$

no relative TFP growth

Final Remarks

- Sectoral influence w/ produced capital = **on impact** + **capital deepening**
  Domar weights depend on the transition path
- Build first **harmonized cross-country measures** of the investment network
- CONS & DUR matter through capital deepening; ICT rises with income through all channels
- Role for income disparities, $\approx 40\%$; heterogeneity in Ω reduces disparities
- Relative prices alone miss the evolution of Ω in some sectors.
- Δ gross return on capital important for ICT and Construction.

Methodology: Example

FROM : ICT



investment production

$100

TO: MAN SER

Methodology: Example

FROM : ICT investment production $100

Employment by occupation

TO:

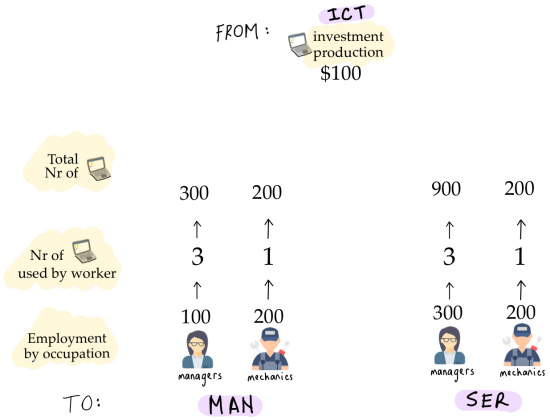


MAN

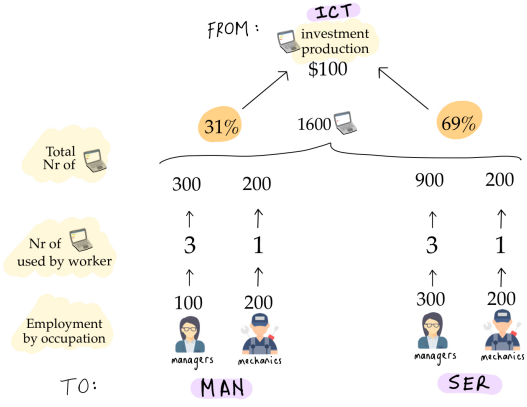


SER

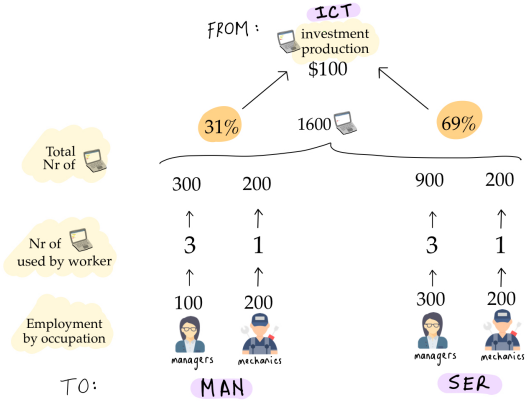
Methodology: Example



Methodology: Example



Methodology: Example



Counterpart in the capital flows table

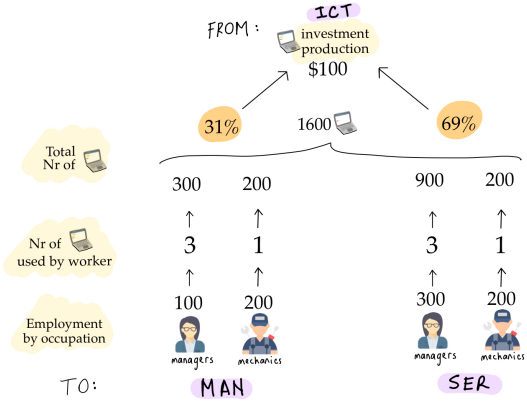
[values in $]

EXPENDITURES

PRODUCTION	AGFi	MAN	SER	Electronics	ICT	Machinery	Transport	Construc
AGFi								
MAN								
SER								
Electronics								
ICT		$31	$69					
Machinery								
Transport								
Construction								

$100

Methodology: Example



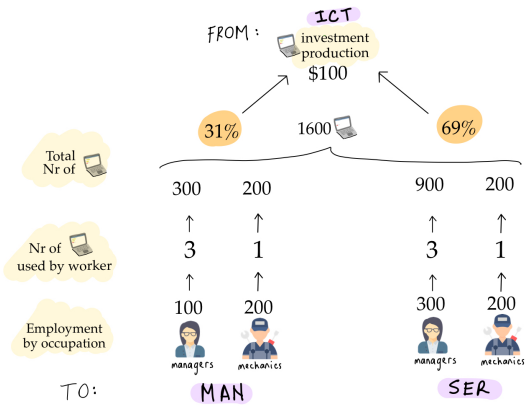
Counterpart in the capital flows table

[values in $]

EXPENDITURES

PRODUCTION	AGRi	MAN	SER	Electronics	ICT	Machinery	Transport	Construction	
AGRi		:	:						
MAN		:	:						
SER		:	:						
Electronics		$20	$50						$70
ICT		$31	$69						$100
Machinery		:	:						
Transport		$80	$30						$110
Construction		:	:						
		$300	$500						

Methodology: Example



Counterpart in the investment network

[values in % share]

EXPENDITURES

PRODUCTION	AGFi	MAN	SER	Electronics	ICT	Machinery	Transport	Construction
AGFi		:	:					
MAN		:	:					
SER		:	:					
Electronics		7%	10%					
ICT		10%	14%					
Machinery		:	:					
Transport		27%	6%					
Construction		:	:					
		100%	100%					

back

Dataset Coverage

Low Income

Country	Years
Ethiopia	1990-2019
Rwanda	1990-2019
Tanzania	1990-2019
Zambia	1990-2019
Kenya	1990-2019
Cambodia	2005-2015
Senegal	1990-2019
India	1965-2014
Vietnam	2005-2015
Ghana	1990-2019
Nigeria	1990-2019
Philippines	2005-2015
Morocco	2005-2015
Tunisia	2005-2015

Middle Income

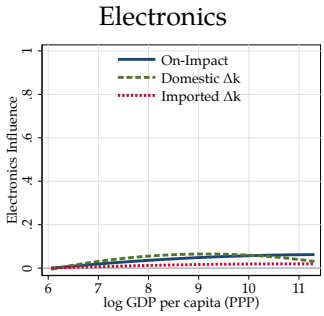
Country	Years
Thailand	2005-2015
South Africa	2005-2015
Costa Rica	2005-2015
Turkey	2000-2014
Argentina	2005-2015
Mauritius	1990-2019
Chile	2005-2015
Mexico	1965-2014
Russia	2000-2014
Poland	2000-2014
Malaysia	2005-2015
Indonesia	2005-2015
Brazil	1965-2014
China	2005-2015
Peru	2005-2015
Colombia	2005-2015

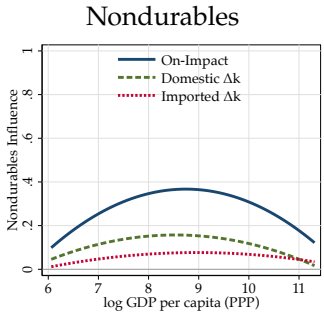
High Income

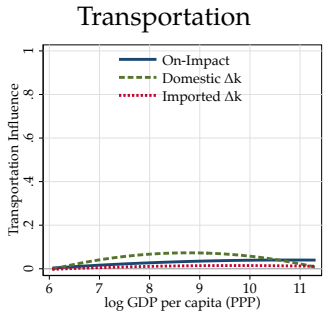
Country	Years
Lithuania	2000-2014
Slovakia	2000-2014
Hungary	2000-2014
Czechia	2000-2014
Portugal	1965-2014
Slovenia	2000-2014
Greece	1965-2014
South Korea	1965-2014
New Zealand	2005-2015
Israel	2005-2015
Spain	1965-2014
Cyprus	2005-2015
Italy	1965-2014
France	1965-2014
Japan	1965-2014
Germany	1965-2014
Belgium	1965-2014
United Kingdom	1965-2014
Denmark	1965-2014
Sweden	1965-2014
Austria	1965-2014
Australia	1965-2014
Netherlands	1965-2014
Ireland	1965-2014
Switzerland	2005-2015
Norway	2000-2014
United States	1965-2014
Singapore	2005-2015

data sources variation sources back

Channels of Sectoral Influence: Other Sectors







back

Development Accounting: Full Table

Exercise	Income Variance
Baseline	1.00
Investment Links Only	0.42
Investment Links Only + US TFP	0.41
Investment Links Only + α avg.	0.62
Domestic (no trade)	1.04
Ω = Identity	1.26
Ω = VA Shares	1.11
Ω = VA Shares, TFP (VA model)	4.23
Baseline Ω, TFP (VA model)	1.16

Data for 2005. "TFP (VA model)": productivity inferred from a value-added model.

back

Standard Development Accounting

Table 5: Development Accounting

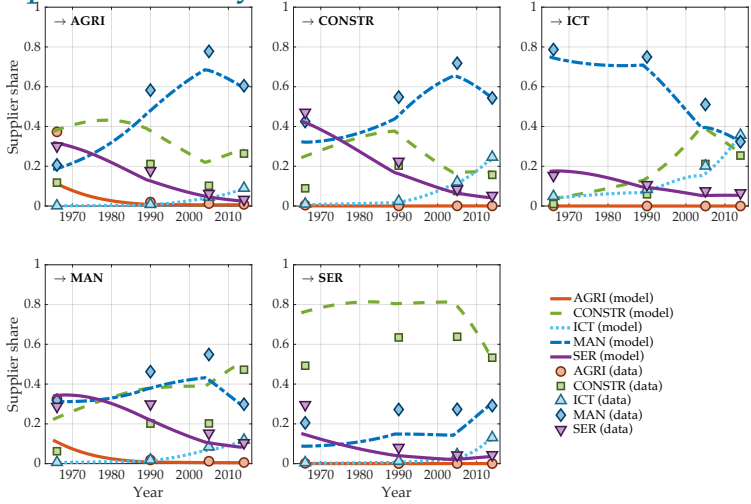
	Income Variance
Caselli (2004) $\frac{var\left[\frac{\alpha}{1-\alpha}\ln\left(\frac{K}{Y}\right)\right]}{var\left[\ln\left(\frac{Y}{L}\right)\right]}$	0.07
Klenow and Rodriguez-Clare (2004) $\frac{var\left[\frac{\alpha}{1-\alpha}\ln\left(\frac{K}{Y}\right)+cov\left(\ln(Z),\ln\left(\frac{K}{Y}\right)\right)\right]}{var\left[\ln\left(\frac{Y}{L}\right)\right]}$	0.2

$$\ln\left(\frac{Y}{L}\right) = \ln(Z) + \frac{\alpha}{1-\alpha}\ln\left(\frac{K}{Y}\right)$$

→ **13%** of GDP p/cap. differences explained by ≠ in aggregate capital stock

back

Investment Input Shares by Destination Sector



Model (lines) vs data (markers). Targeted: odds ratios of the network & sectoral composition of aggregate investment.

back

Calibration: Transition Dynamics

Parameter	Description	Value	Moment/Calibration
Sector order for vectors: Agriculture, Construction, ICT, Manufacturing, Services			
Technology			
γ_i	VA share in gross output	{0.58, 0.35, 0.46, 0.25, 0.57}	Sectoral VA/gross-output ratios
α_i	Capital share in VA	{0.09, 0.15, 0.42, 0.55, 0.31}	Sectoral capital income shares
δ	Depreciation rate	0.10	Standard
ι	Elast. of subst., inv.	2.41	Regression of investment shares on prices
$\tilde{\omega}_{in,t}$	Investment weights	see panel	Sectoral investment shares (aggregate) Investment-network odds ratios
$\mu_{in,t}$	IO network	see panel	Input–output network
$z_{j,t}$	Sectoral productivities	see figure	Relative prices; GDP per capita, 2014/1965
Preferences			
β	Discount factor	0.96	4% interest rate
σ	Elast. of subst., cons.	0.5	Consumption goods complements
ϵ_j	Income elasticities	{0.20, 1.00, 1.20, 1.00, 0.80}	Income variation of consumption shares
θ_j	Consumption weights	{0.07, 0.00, 0.07, 0.29, 0.57}	Average consumption shares, 1965–2014
Wedges			
τ_t	Euler-equation wedge: $(1+\tau_t)\frac{p^x_{n,t}}{p^x_{n,t+1}} = \beta\frac{U'(C_{t+1})}{U'(C_t)}\left(\frac{R_{n,t+1}}{p^x_{n,t+1}}+1-\delta\right)$	time-varying	Aggregate investment share $I/(C+I)$
τ^{ℓ}_i	Sectoral labor wedges	{0.06, −0.07, −0.04, 0.00, 0.06}	Sectoral wage gaps

$\tilde{\omega}$ panel | μ panel | productivity paths | back

Calibration: Investment Weights $\tilde{\omega}_{in,t}$

	1966					1990					2005					2014				
Source i	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.
Agri.	0.18	0.00	0.00	0.14	0.00	0.06	0.00	0.00	0.05	0.00	0.06	0.00	0.00	0.04	0.00	0.06	0.00	0.00	0.03	0.00
Cons.	0.19	0.11	0.01	0.09	0.56	0.33	0.33	0.12	0.30	0.78	0.35	0.25	0.49	0.43	0.87	0.50	0.30	0.38	0.60	0.71
ICT	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.06	0.01	0.00	0.06	0.17	0.20	0.08	0.03	0.13	0.35	0.39	0.11	0.13
Manuf.	0.42	0.64	0.92	0.58	0.29	0.37	0.35	0.63	0.27	0.13	0.33	0.31	0.15	0.15	0.05	0.22	0.20	0.09	0.07	0.08
Serv.	0.21	0.25	0.06	0.18	0.15	0.24	0.31	0.19	0.38	0.08	0.20	0.26	0.16	0.30	0.05	0.09	0.15	0.14	0.19	0.08

Columns are destination sectors and sum to one. Recovered to match the investment-network odds ratios and the sectoral composition of aggregate investment at each anchor ($\iota = 2.41$). Interpolated log-linearly between anchors, flat after 2014.

back

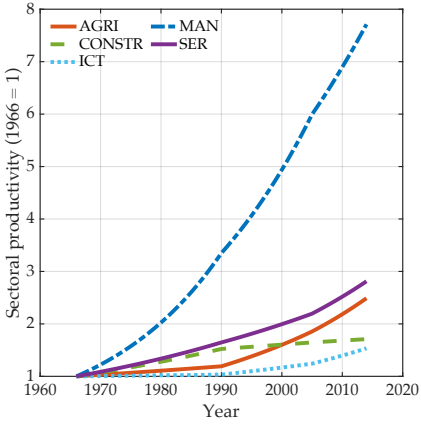
Calibration: Input–Output Network $\mu_{in,t}$

	1966					1990					2005					2014				
Source i	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.	Agri.	Cons.	ICT	Manuf.	Serv.
Agri.	0.50	0.01	0.00	0.40	0.01	0.23	0.00	0.00	0.10	0.00	0.16	0.01	0.00	0.04	0.01	0.16	0.01	0.00	0.03	0.01
Cons.	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.02
ICT	0.00	0.03	0.07	0.01	0.07	0.01	0.08	0.16	0.04	0.13	0.02	0.12	0.54	0.07	0.15	0.01	0.12	0.38	0.03	0.17
Manuf.	0.37	0.58	0.63	0.48	0.23	0.54	0.62	0.58	0.67	0.25	0.62	0.69	0.20	0.75	0.28	0.66	0.68	0.27	0.75	0.33
Serv.	0.12	0.37	0.30	0.11	0.66	0.22	0.29	0.26	0.19	0.57	0.20	0.18	0.26	0.14	0.54	0.17	0.18	0.35	0.18	0.48

Columns are destination sectors and sum to one. Source: WIOD. Interpolated linearly between anchors, flat after 2014.

back

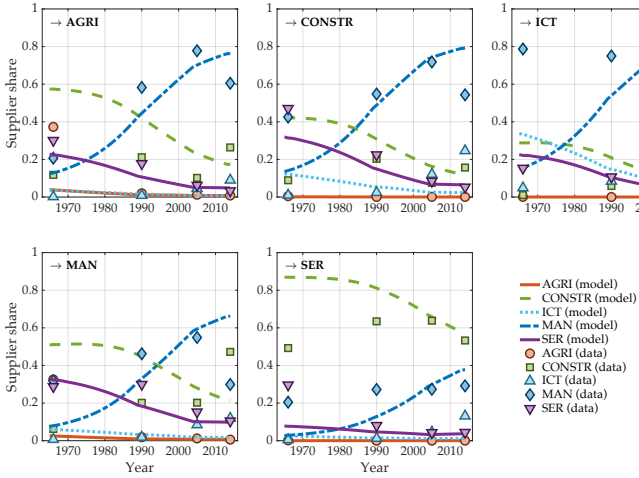
Calibration: Sectoral Productivity Paths



$z_{j,t}$ indexed to one in 1966. MAN rises roughly eightfold; other sectors between 1.5- and 3-fold. Log-linear between anchors 1966/1990/2005/2014.

back

Investment Network by Destination: Constant $\tilde{\omega}$

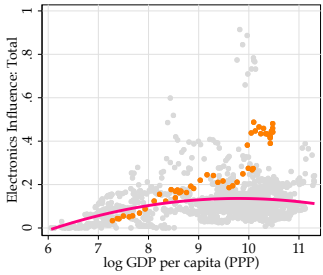


Model (lines) vs data (markers), $\tilde{\omega}$ held constant so Ω moves only through relative prices.

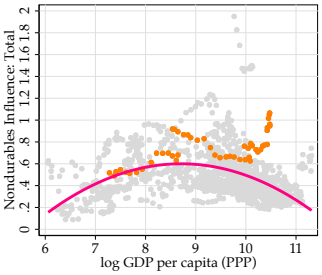
back

Sectorial Influence: Other Sectors

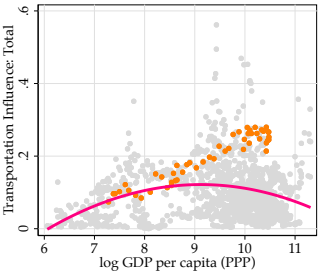
Electronics



Nondurables



Transportation



back

Aggregate Growth Path

Proposition (AGP)

CD value added, CES aggregators from $\bar{t}$ on, investment inputs all substitutes (or all complements): asymptotic AGP where GDP and wages grow at constant rates. One sector n dominates investment; one sector n^* dominates final expenditure.

$$\hat{g}_w = \frac{(1-\phi_n)(\mathcal{A}g_z)_n + \phi_n \kappa_{n^*}(\mathcal{A}g_z)_{n^*}}{(1-\phi_n)\mathcal{B}_n + \phi_n\left[1-\kappa_{n^*}(1-\mathcal{B}_{n^*})\right]} \qquad g_Y = \kappa_{n^*}\left[(1-\mathcal{B}_{n^*})\hat{g}_w + (\mathcal{A}g_z)_{n^*}\right]$$

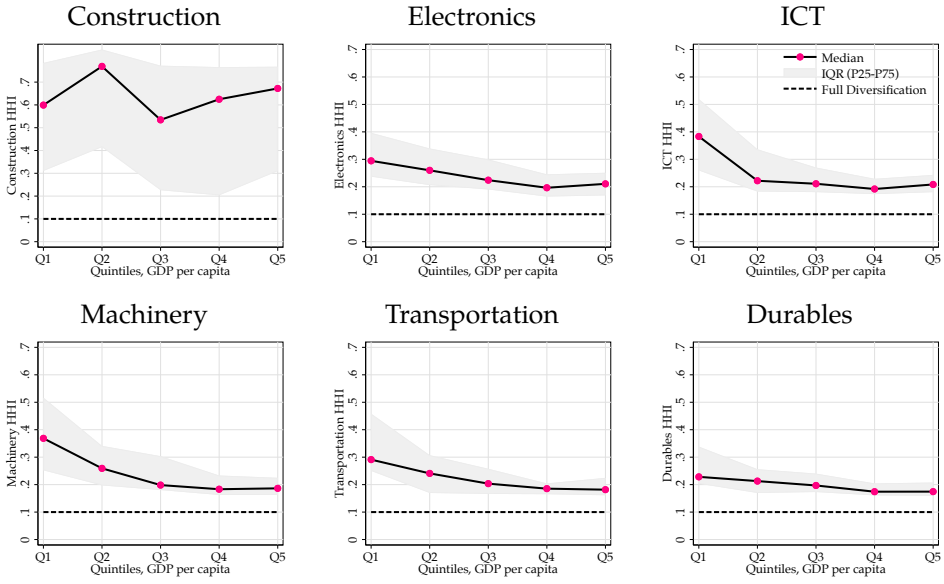
$$g_{p_j^x} = g_{R_j} = g_Y - \hat{g}_w, \qquad g_{k_j} = \hat{g}_w + g_{l_j}$$

$\hat{g}_w \equiv g_W - g_{p_n^x}$; $\mathcal{A} \equiv \Psi\Gamma^\infty$, $\mathcal{B} \equiv \mathcal{A}(I-\alpha)\mathbf{1}$ (embodied labor share), Ψ Leontief inverse, $\kappa_{n^*} \equiv \frac{\sigma-1}{\sigma-\epsilon_{n^*}}$

closed economy: $\hat{g}_w = \frac{(\mathcal{A}g_z)_n}{\mathcal{B}_n}$ (1 sector: $g_z/(1-\alpha)$)

back

Investment network: HHI demand for investment



back

Welfare

Proposition (Welfare)

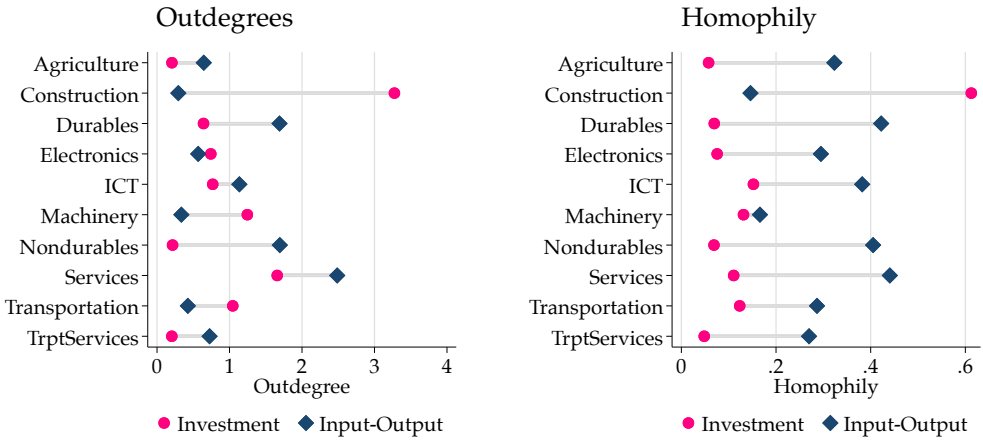
$$\textit{Welfare influence} \equiv \frac{d \ln C_t}{d \ln z_{nt}} = \underbrace{\overbrace{\frac{p_t v_t}{C_t}}^{GDP}}_{cons.} \tilde{\eta}_{nt} \gamma_{nt}$$

$$\tilde{\eta}_t = \underbrace{[I - (I - \phi)\Omega_t \Gamma_t \alpha_t \rho_t - M_t(I - \Gamma_t)]^{-1}}_{augmented\ Leontief\ inverse} \underbrace{\tilde{\zeta}_t}_{final\ uses}$$

envelope theorem; GDP/C constant along AGP → amplified or dampened along the transition

back

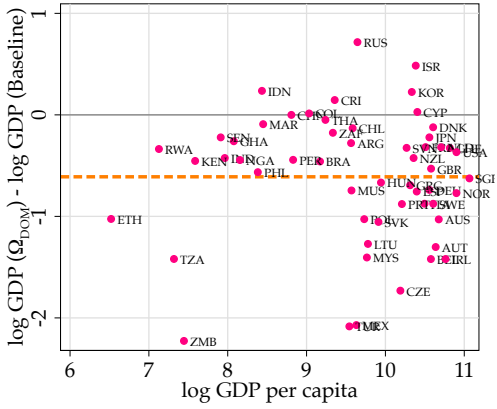
Investment Network vs. Input–Output



Data for 2005, averaged across countries. Circles = investment network; diamonds = input–output network.

back

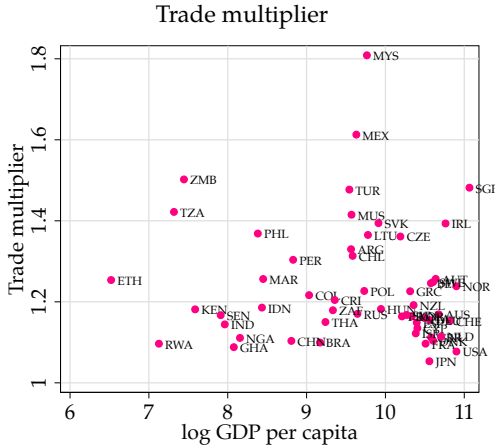
Development Accounting: Income Levels



...and average income per capita levels falls by 60%

back

Development Accounting: the Impact of Trade



back

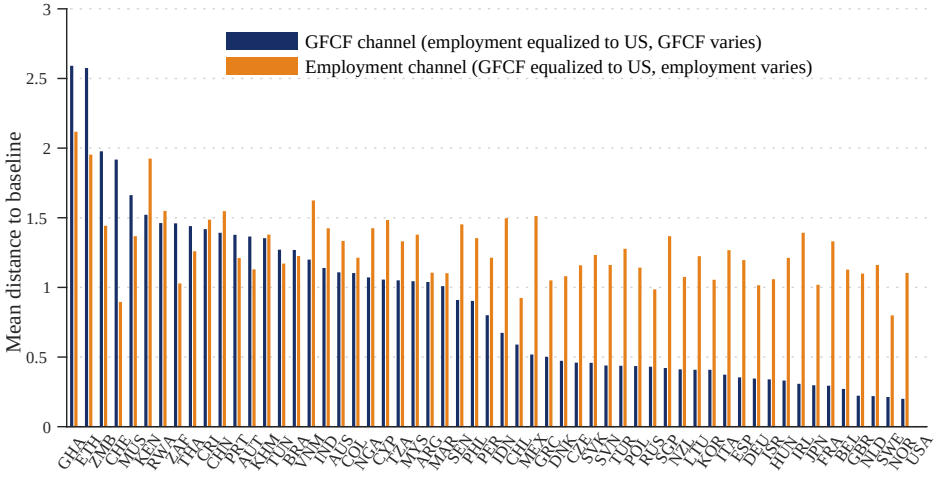
Investment Network: Data Sources

Data description	Source
Sector–equipment bridge	US Bridge Tables (BEA)
Number of tools per worker by occupation	Caunedo et al. (2023)
Investment production by sector	WIOD; OECD; Mensah & de Vries (2023)
Input–output structure	WIOD; OECD; Mensah & de Vries (2023)
Employment by occupation and sector	IPUMS; ILOSTAT; PIAAC

WIOD = World Input–Output Database; OECD = Organisation for Economic Co-operation and Development; BEA = U.S. Bureau of Economic Analysis; IPUMS = Integrated Public Use Microdata Series; ILOSTAT = ILO Statistics; PIAAC = Programme for the International Assessment of Adult Competencies.

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Within-Country Distance: Baseline vs Counterfactual



Each bar reports the mean Frobenius distance between the baseline investment network and a counterfactual, averaged over all available years.

back

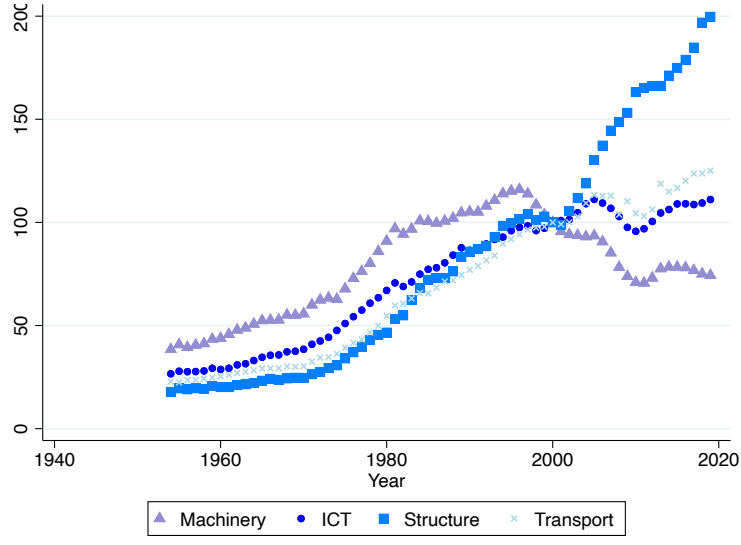
Sectoral Influence

Sector	Low income	Middle income	High income
Agriculture	0.38 (0.28 0.51)	0.15 (0.08 0.20)	0.04 (0.02 0.06)
Construction	0.17 (0.07 0.26)	0.25 (0.21 0.33)	0.19 (0.13 0.21)
Durables	0.13 (0.05 0.18)	0.20 (0.13 0.21)	0.13 (0.10 0.16)
Electronics	0.05 (0.01 0.05)	0.10 (0.05 0.10)	0.06 (0.04 0.07)
ICT	0.14 (0.10 0.15)	0.23 (0.16 0.29)	0.31 (0.25 0.37)
Machinery	0.03 (0.01 0.04)	0.07 (0.05 0.08)	0.04 (0.03 0.06)
Nondurables	0.20 (0.10 0.31)	0.34 (0.23 0.44)	0.13 (0.09 0.15)
Services	0.67 (0.59 0.80)	0.69 (0.51 0.84)	0.75 (0.69 0.81)
Transportation	0.03 (0.01 0.05)	0.07 (0.05 0.08)	0.04 (0.03 0.06)
TrptServices	0.10 (0.07 0.13)	0.14 (0.09 0.15)	0.11 (0.08 0.12)

Notes: Data for 2005. Entries report means; values in parentheses are the 25th and 75th percentiles.

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Price of Capital, US, (P_{2000}=100)



back

Comparison with vom Lehn & Winberry (2022), US 2012

Outdegrees

Sector	This Paper	VLW
Agriculture	0.00	0.00
Construction	1.65	1.31
Durables	0.30	0.24
Electronics	0.90	0.64
ICT	3.02	3.57
Machinery	1.42	1.94
Nondurables	0.04	0.03
Services	1.16	1.06
Transportation	1.35	1.10
Trpt Services	0.16	0.10

Homophily

Sector	This Paper	VLW
Agriculture	0.00	0.00
Construction	0.04	0.03
Durables	0.04	0.05
Electronics	0.12	0.07
ICT	0.43	0.61
Machinery	0.18	0.26
Nondurables	0.01	0.005
Services	0.11	0.12
Transportation	0.11	0.06
Trpt Services	0.03	0.01

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Validation: Chile, firm-to-firm VAT data

only domestic transactions

Outdegrees

Sector	This Paper	GHKQR
Agriculture	0.32	0.07
Construction	4.22	5.13
Durables	0.44	0.46
Electronics	0.06	0.11
ICT	2.73	0.15
Machinery	0.49	0.28
Nondurables	0.34	0.15
Services	1.08	4.47
Transportation	0.13	0.04
Trpt Services	0.20	0.11

Homophily

Sector	This Paper	GHKQR
Agriculture	0.13	0.04
Construction	0.68	0.86
Durables	0.07	0.07
Electronics	0.01	0.05
ICT	0.33	0.03
Machinery	0.07	0.11
Nondurables	0.09	0.04
Services	0.06	0.36
Transportation	0.01	0.01
Trpt Services	0.06	0.04

GHKQR = Gillmore et al. (2025)

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Investment Network Outdegrees & Development

Sector	Low income	Middle income	High income
Agriculture	0.45 (0.06 0.48)	0.22 (0.05 0.32)	0.13 (0.05 0.21)
Construction	3.18 (2.69 3.89)	3.39 (3.07 3.63)	2.98 (2.55 3.35)
Durables	0.90 (0.24 0.97)	0.39 (0.31 0.48)	0.49 (0.30 0.70)
Electronics	0.69 (0.20 0.96)	0.97 (0.64 1.13)	0.78 (0.57 0.95)
ICT	0.41 (0.01 0.90)	0.46 (0.28 0.58)	1.21 (0.84 1.38)
Machinery	1.42 (0.36 1.80)	1.31 (1.08 1.68)	1.29 (0.99 1.64)
Nondurables	0.32 (0.08 0.43)	0.16 (0.05 0.26)	0.18 (0.14 0.22)
Services	1.75 (1.26 2.38)	1.61 (1.28 1.76)	1.58 (1.16 1.95)
Transportation	0.72 (0.55 1.01)	1.08 (0.92 1.24)	1.20 (0.84 1.53)
TrptServices	0.14 (0.00 0.12)	0.41 (0.15 0.44)	0.16 (0.08 0.18)

Notes: Data for 2005. Outdegrees are sectoral row sums in the investment network. Entries report means; values in parentheses are the 25th and 75th percentiles.

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Counterfactual: No Relative TFP Growth, One Sector at a Time

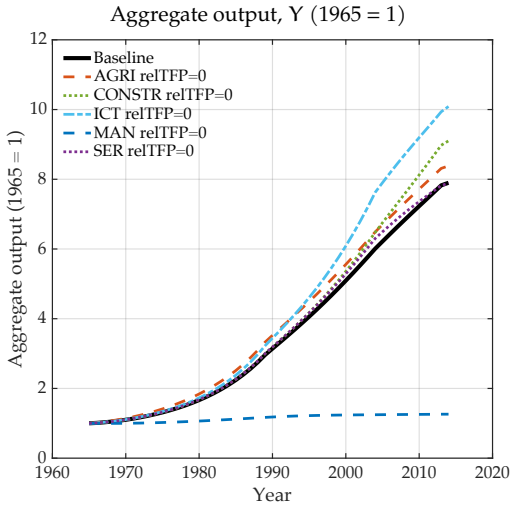
Frozen sector	$\Delta \ln Y$	$\Delta \ln(K/Y)$	$\frac{\Delta \ln Y}{\Delta \ln(K/Y)}$	$\Delta \ln Y - \frac{\bar{\alpha}}{1-\bar{\alpha}} \Delta \ln(K/Y)$	productivity % of $\Delta \ln Y$
Baseline	2.04	0.88	2.31	1.66	81%
Agriculture	2.10	0.85	2.47	1.73	82%
Construction	2.17	1.01	2.15	1.73	80%
ICT	2.28	0.95	2.39	1.87	82%
Manufacturing	0.23	0.91	0.25	−0.17	−73%
Services	2.04	0.91	2.25	1.64	81%

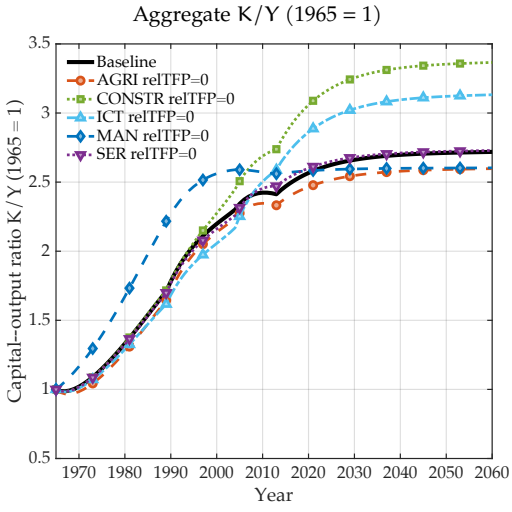
Detrended log changes 1965–2014, chained Törnqvist Y and K. Each row freezes one sector's MAN-relative TFP at its 1966 level. $\bar{\alpha} = 0.30$: simple average of sectoral capital shares, $\frac{\bar{\alpha}}{1-\bar{\alpha}} = 0.44$.

→ **MAN productivity** is the key driver of Korea's growth miracle

output & K/Y paths | sectoral capital | back

No Relative TFP Growth: Output and Capital–Output Ratio

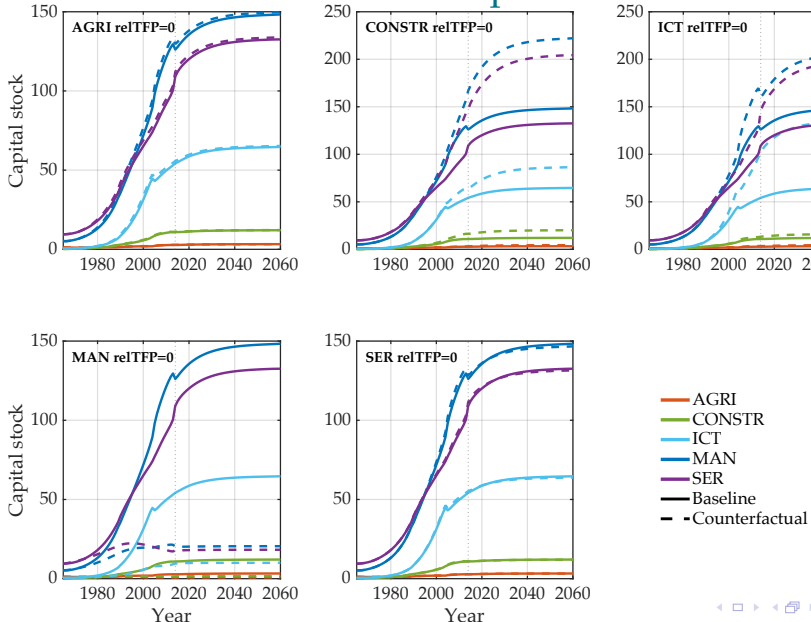




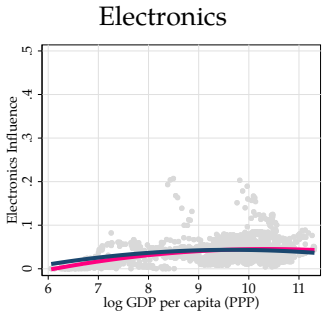
Detrended, chained Törnqvist indices. Each line freezes one sector's MAN-relative TFP at its 1966 level.

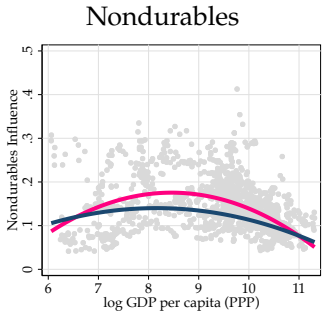
sectoral capital back

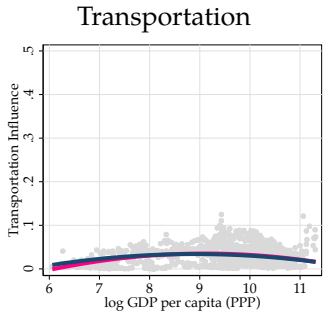
No Relative TFP Growth: Sectoral Capital



Sectorial Influence: VA shares fixed USA 2014 (other sectors)

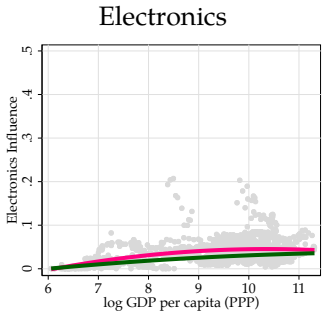


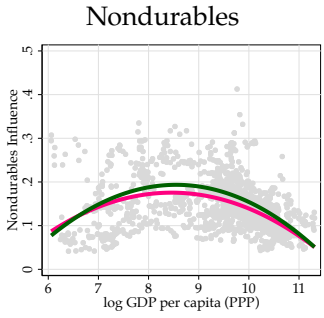


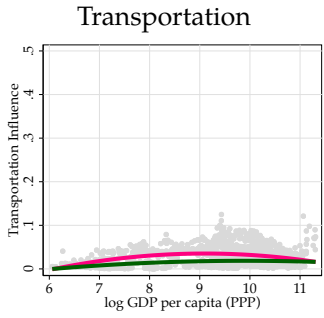


back

Sectorial Influence: INV cols = VA shares (other sectors)

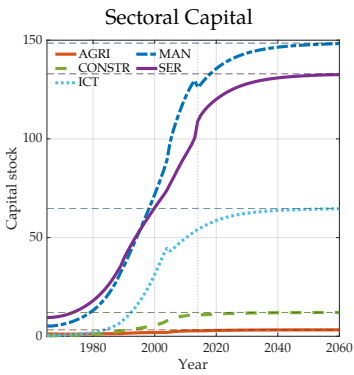


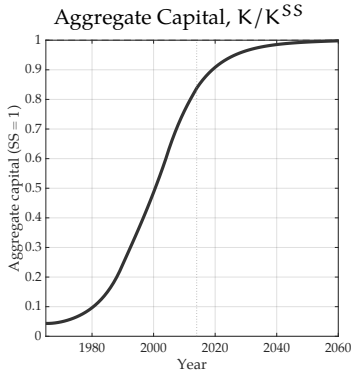


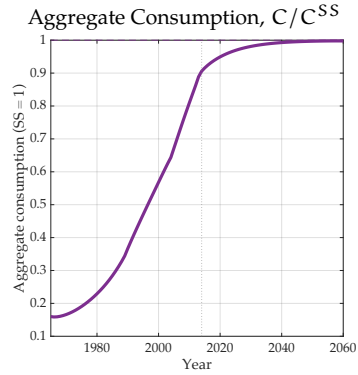


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Transition Paths: Levels







- 1965–2014: MAN capital ×24, SERV ×12, ICT > ×100; AGRI barely doubles
- Aggregate K: 4% → 84% of SS; aggregate C: 16% → 90% of SS

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The Model: trade

- Investment services: domestic + imported components

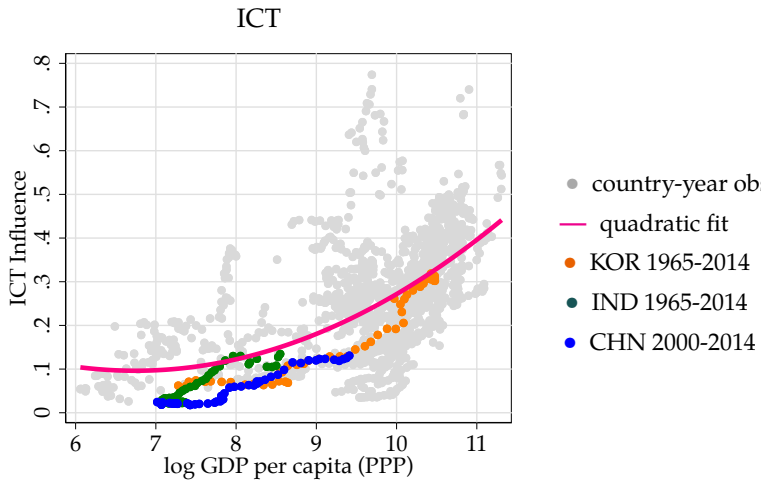
$$\chi_{int} = \underbrace{\left(\frac{\chi^{d}_{int}}{1-\phi_n}\right)^{1-\phi_n}}_{\text{domestic}} \underbrace{\left(\frac{\chi^{f}_{int}}{\phi_n}\right)^{\phi_n}}_{\text{imported}}$$

- Aggregate trade balance

$$p_Y \epsilon_t - p^f_x \epsilon^f_t = 0$$

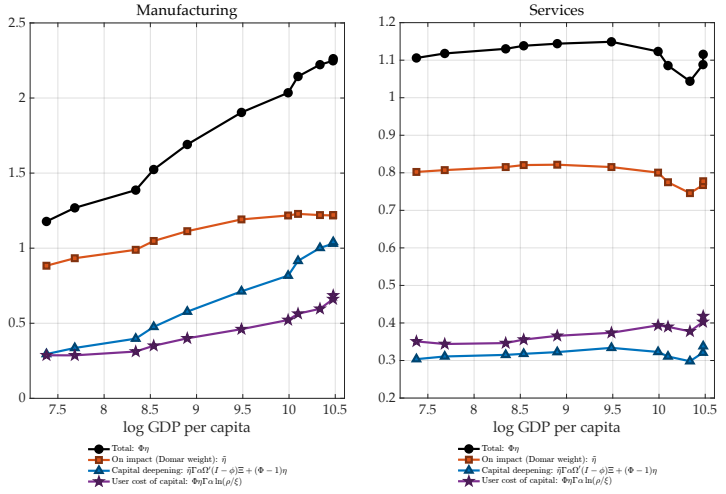
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Sectorial Influence: $\frac{\partial \ln(GDP)}{\partial \ln(a_s)}$



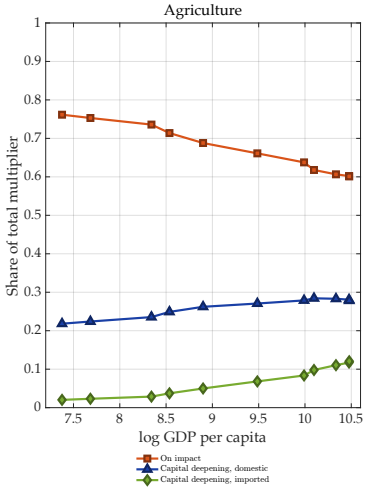
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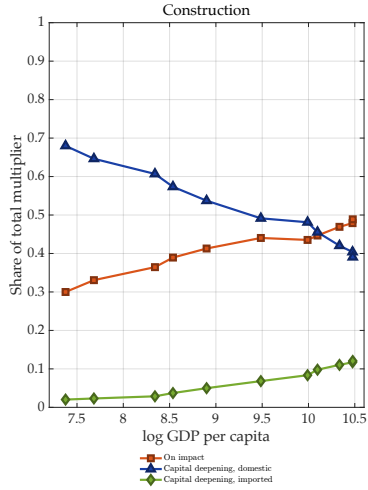
Influence Channels along Korea's Transition: MAN & SERV

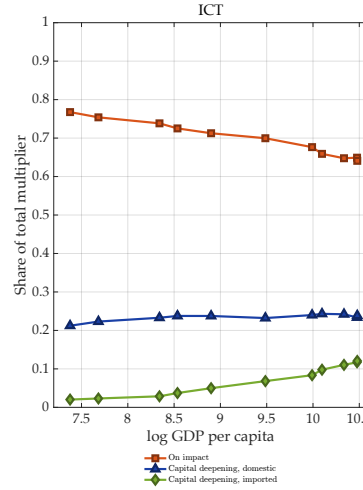


shares of total back

Influence Channels, Korea: Shares of Total

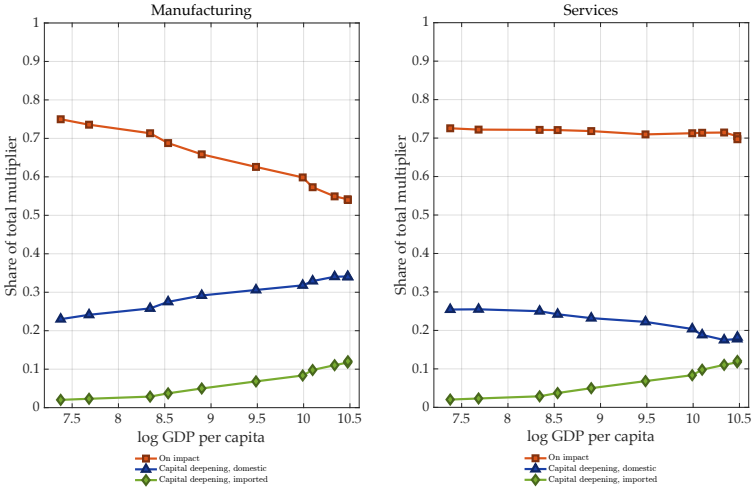






back

Influence Channels, Korea: Shares of Total, MAN & SERV



back